

# Analysis of unannotated expressed regions in transcriptomic data from *Mycobacterium Abscessus*

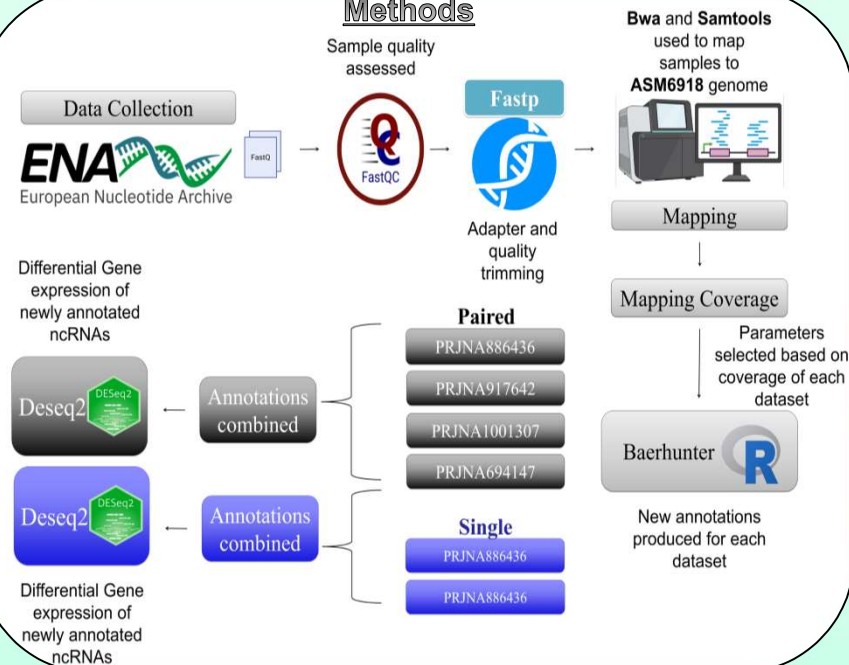


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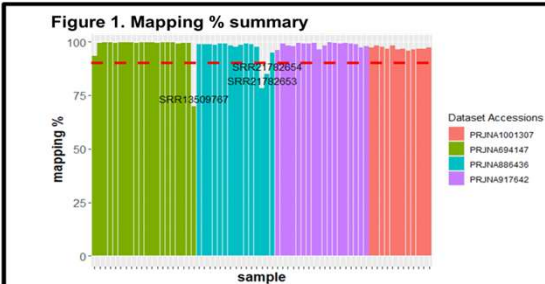
## Introduction

*Mycobacterium abscessus* is a highly **antibiotic-resistant pathogen** that poses a serious threat to vulnerable populations, particularly individuals with cystic fibrosis. The annotation of **non-coding RNAs (ncRNAs)** in mycobacteria is particularly challenging, as these elements often display little sequence similarity to those found in other taxa. However, leveraging **expression-based annotation** methods offers a powerful advantage by helping to identify the most transcriptionally active and potentially **functionally relevant genomic regions**. This approach is especially important as they can help **modulate gene expression** facilitating rapid adaptability to the environment. Recent discoveries surrounding ncRNAs in model organisms and their involvement in **pathogenicity and virulence** further emphasising the importance of studying these in other bacterial species. Notably, the current genome assembly contains only **42 annotated ncRNAs**, highlighting the need for use of such strategies to detect novel ncRNAs

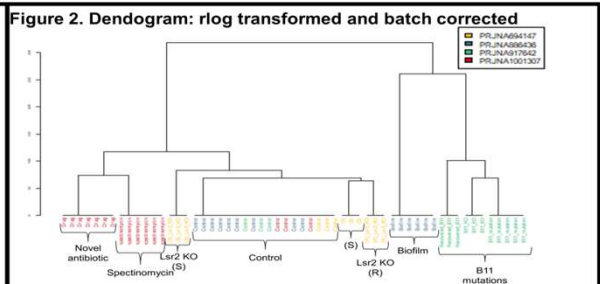
## Methods



## Results



Mapping percentage was in high 90s for all samples excluding three samples highlighted in figure 1 (SRR21782654, SRR21782653, SRR13509767) which fell below 90% and therefore removed from further analysis.



Following log transformation and batch correction samples grouped by condition rather than dataset meaning any differences in expression can be attributed to differing conditions and not just difference in sequencing between datasets.

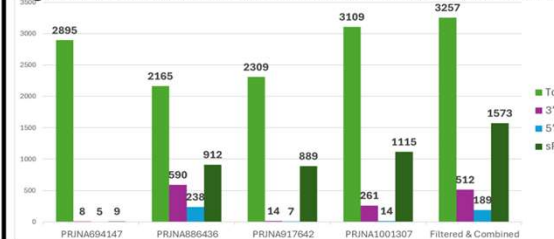
Table 1: Median mapping Coverage Of Intergenic Reads

|              | P25 | P50 | P75 | P80 | P85 | P90 |
|--------------|-----|-----|-----|-----|-----|-----|
| PRJNA694147  | 4   | 11  | 30  | 38  | 51  | 76  |
| PRJNA886436  | 32  | 67  | 156 | 195 | 264 | 377 |
| PRJNA917642  | 6   | 17  | 46  | 60  | 82  | 124 |
| PRJNA1001307 | 11  | 41  | 129 | 169 | 230 | 339 |

Baerhunter initial Coverage parameters

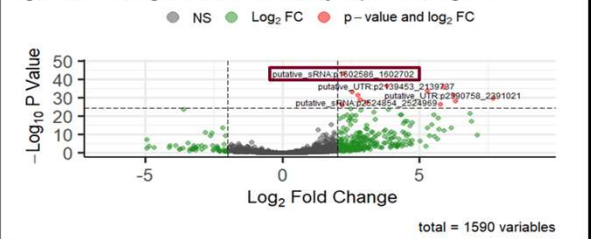
Coverage of intergenic reads was calculated, median 90th Percentile of mapping coverage for all the samples in each dataset was used as a starting point for Baerhunter coverage parameters. Both "high coverage\_cutoff" and "low\_coverage\_cutoff" were set to P90 and adjusted from there through trial and error to achieve sufficient sensitivity without excessive background noise.

Figure 3: Baerhunter Annotations Produced From Each Dataset



Putative ncRNAs for each dataset were predicted using Baerhunter. Putative features were further processed to resolve overlaps and filtered based on expression (TPM>1000) and length (<1000) to create a combined file. Total identified ncRNAs were then recounted against the combined annotations.

Figure 4: Most significant differentially expressed regions



Differential analysis utilising Deseq2 package showed a total of 679 (28%) of features as upregulated and 336 (14%) were downregulated. While 522 of the upregulated features and 129 of the downregulated features were putative ncRNAs. Most significantly differentially expressed was a putative antisense sRNA outlined found on the opposite strand of for Aminopeptidase N gene and is likely key to its regulation.

## Discussion and Conclusions

High mapping percentages across most samples support dataset reliability. Baerhunter, guided by intergenic coverage metrics and refined iteratively, enabled sensitive ncRNA prediction, though further **parameter optimisation** is needed to reduce background noise. Additionally, completing the analysis of single end datasets will hopefully provide further data to support current findings. The rlog transformation and batch correction clarified that **expression differences stemmed from biological conditions** rather than technical variation. Of the differentially expressed features, **651 were putative ncRNAs**, demonstrating condition-specific transcriptional activity. This suggests unannotated ncRNAs play a key role in *M. abscessus* adaptation. While **plasmid features were excluded**, they may represent additional regulatory complexity. These results highlight widespread regulation of unannotated ncRNAs and the need for **experimental validation** and **functional characterization** to confirm their role in environmental responsiveness. Promising candidates should be identified for further investigation, along with an assessment of potential downstream effects of expressed transcript.

## References

1. Stiens, Jennifer et al., 2023. *Molecular Microbiology* 119 (4): 381–400.
2. Alina Ozuna et al., 2019. *Bioinformatics* 36 (3): 966–69.
3. Stiens, Jennifer et al., *Molecular Microbiology*. 2021;117(1):20-31.
4. Sivasankaran, A. [Unpublished master's dissertation]. Birkbeck College, University of London.